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Machine learning for phase separation order parameters by using graph networks[†]Yaqiang Zhang^a, Yinhan Zhang^a, Xiwen Wang^a, Yanfang Lv^a and Wen Zheng^{*a,b}

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Phase separation, the process of spontaneous separation of components of multi-component mixtures, is a key aspect of many physical, chemical and biological systems. In this work, we used dissipative particle dynamics to implement the process of phase separation. We designed an order parameter based on local density to measure the evolution of phase separation, and simulated the phase separation process under different repulsion coefficient and different Schmidt numbers. We show that the order parameter can accurately describe the phase separation in different states. Meanwhile, we propose a graph network-based data-driven approach to accurately predict the evolution of the ordinal covariance and the long-term evolution of the particle positions of the phase separation system from the position and species of the particles at the initial moment only, combined with the conserved forces and relative distances of the particle interactions as features to learn the changes of the droplet aggregation during the simulation.

1 Introduction

From macroscopic phase separation of water and oil to microscopic phase separation of polymeric block copolymer materials^{1–7}, phase separation phenomena are ubiquitous. Previously, the evolutionary processes used to understand phase separation were performed by many simulation methods, including microscopic molecular dynamics (MD); Lattice Gas Automata (LGA); dissipative particle dynamics and Navier-Stokes (NS) equations. In recent years, dissipative particle dynamics (DPD) as a coarse-grained simulation method, where the forces between particles take into account the action potential and the random motion of particles, satisfying Newton's second law, has led to the wide application of DPD methods for fluid phase separation and complex fluid simulation. The phase separation process of complex solutions can be directly simulated based on particle systems and suitable force fields^{8,9}.

The influence of the properties of different types of particles on the phase separation of binary mixtures is very important, especially the variation in viscosity due to the different repulsion and friction coefficients^{10–12}. In multiphase systems, the repulsion coefficients between different types of fluids determine the separation strength of phase separation. Exploring the effect of different sizes of repulsion coefficients in the phase separation process has also been a hot topic of research. Meanwhile, each fluid has its own friction factor, which gives rise to the viscosity of the fluid,

and the difference in viscosity and viscosity between one fluid and another has a great influence on the kinetic behavior. So it is very interesting to explore the hydrodynamic changes of the liquid due to the variation of the Schmidt number. In the simulated system, the particles move with time variation and exhibit different characteristics at the macroscopic level. Therefore, the spatial distribution formed based on the positions of different particles is extremely important, for which we designed an ordinal covariance based on the spatial density distribution to dynamically characterize the evolution of phase separation.

It is now well-established that the growth of domains during the phase separation is a scaling phenomenon, e.g., the two-point equal-time correlation function $C(r, t)$ and its Fourier transform, the structure factor $S(k, t)$, characterizing the domain morphology and growth. The spontaneous phase separation is described by a power law of the form $R(t) \sim t^\beta$, where $R(t)$ is the domain size at the time instant t . The exponent β referred to as the growth exponent varies according to the mechanism driving the phase separation process. However, for non-equal volume fraction mixtures, β depends strongly on the volume fraction of the component phases. We want to define a new order parameter to effectively describe the process of phase separation.

Phase separation, due to its importance, has been extensively studied both experimentally and theoretically, but resolving the temporal and spatial variations of molecular dynamics simulations in these complex systems is still a challenging task. Therefore, the use of information technology for their simulation is particularly crucial.

Machine learning methods have recently shown great potential in the analysis of physical systems, where the flexibility

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of neural network models allows computers to learn properties unique to complex physical systems from simple data representations^{13–24}, while learning important features of the data distribution. Sanchez-Gonzalez et al. used a graph-based simulator to achieve long-time predictions for physical systems such as fluids and rigid solids²⁵. Bapst et al. have shown that graph neural networks can obtain long-term kinetic characteristics from the complex structure of liquids²⁶. Considering the potential forces between pairs of particles, Shiba et al. obtained more accurate results in liquid simulations by simultaneously learning the motion tendencies of pairs of particles through graph neural networks²⁷. Farimani et al, correctly predicted the occurrence of phase transition by conditional generative adversarial networks used for learning, modeling and prediction of phase separation²⁸. The results demonstrate that graph neural networks show very high predictive power in millions of molecular dynamics simulation steps, which makes the combination of machine learning and phase separation simulation a new research hotspot.

In this work, we combine phase separation simulations and machine learning, using dissipative particle dynamics to accomplish the simulation of phase separation. We propose a local density-based order parameter to measure the evolution of phase separation and discuss the phase separation process under different repulsion coefficients and symmetric and asymmetric Schmidt numbers. In addition to this, a GNN model is designed for simulating the phase separation. The results demonstrate that our GNN model can learn the distance variation of paired particles and the conserved forces between paired particles from the particle positions and particle species at the initial moment, capturing the physical processes behind the complex dynamics.

2 Simulation methods and details

2.1 Dissipative particle dynamics

Dissipative particle dynamics (DPD) is mainly used to simulate the behavior of complex fluids, with the advantage that particle coarsening and the use of soft potentials allow a large improvement in the time and space scales of the simulated system and a high level of accuracy. Where a single particle represents a molecular or atomic cluster and the force between pairs of particles consists of three paired components²⁹.

$$\mathbf{F}_i = \sum_{j \neq i} (\mathbf{F}_{ij}^C + \mathbf{F}_{ij}^D + \mathbf{F}_{ij}^R) \quad (1)$$

where $\mathbf{F}_{ij}^C$ is the conservative force, $\mathbf{F}_{ij}^D$ is the dissipative force, $\mathbf{F}_{ij}^R$ is the random force. These relations are defined as

$$\begin{aligned} \mathbf{F}_{ij}^C &= a_{ij} \omega^C(r_{ij}) \hat{\mathbf{r}}_{ij} \\ \mathbf{F}_{ij}^D &= -\gamma \omega^D(r_{ij}) (\hat{\mathbf{r}}_{ij} \cdot \mathbf{v}_{ij}) \hat{\mathbf{r}}_{ij} \\ \mathbf{F}_{ij}^R &= \sigma \omega^R(r_{ij}) \xi_{ij} \Delta t^{-1/2} \hat{\mathbf{r}}_{ij} \end{aligned} \quad (2)$$

where a_{ij} is the repulsion coefficient between particle i and particle j , and $\mathbf{r}_i$, $\mathbf{r}_j$ is the position vector of particle i and particle j , $\mathbf{v}_{ij}$ is the relative velocity of the two particles, $\mathbf{r}_{ij} = \mathbf{r}_i - \mathbf{r}_j$, $r_{ij} = |\mathbf{r}_{ij}|$.

γ and σ are the coefficient of friction and the strength of the random force, respectively, Δt is the time step, ξ_{ij} denotes a fluctuating variable with a mean of 0 and a random nature³⁰. $\omega^C(r_{ij})$, $\omega^D(r_{ij})$ and $\omega^R(r_{ij})$ are the weight functions of the conservative, dissipative and random forces, respectively. These weighting functions are defined by the following relations.

$$\begin{aligned} \omega^C(r_{ij}) &= 1 - \frac{r_{ij}}{r_c}, \quad r_{ij} < r_c \\ \omega^D(r_{ij}) &= \omega^R(r_{ij}) = (1 - \frac{r_{ij}}{r_c})^2, \quad r_{ij} < r_c \end{aligned} \quad (3)$$

r_c is the cutoff radius, σ and γ are set up to satisfy the rise and fall dissipation theorem $\sigma^2 = 2k_B T \gamma$, k_B is the Boltzmann constant, T denotes the system temperature.

2.2 Simulation parameters and data set settings

Here, we choose to simulate binary mixtures in two dimensions for different types of fluids by varying the repulsion coefficients between two different types of particles. The size of our simulation box is $L_x \times L_y = 20 \times 20$ with periodic boundary conditions. The density of the particles in the box is set to $\rho = 3$, the number of each of the two types of particles in the system is $N_A = N_B = 600$, so the total number of all particles is $N = N_A + N_B = 1200$. We use the dimensionless value of the cutoff radius (r_c) = 1, the temperature of the system is set to 1, mass of the particle ($m_A = m_B$) = 1, these settings are reasonable choices for DPD simulated fluids. To achieve two-phase separation, we set the repulsive interaction coefficient between any two particles of the same component to be set to $a_{AB} = 25$, and the repulsive interaction coefficient between two particles of different components to be set to $a_{AB} = 50$.

The viscosity of the DPD fluid can be controlled by the friction coefficient, which is an input parameter in DPD that characterizes the strength of the resistance between interacting particles¹². We achieve a change in the viscosity of the system by varying the friction coefficient (γ), and Table 1 indicates the values of the dissipation coefficients considered in this work and the corresponding Schmidt numbers (Sc). The friction coefficients between the different species of particles are the harmonic average of their respective friction coefficients.

Table 1 The Schmidt number Sc corresponding to different friction coefficients γ .

γ	Sc
4.5	1
22.5	2.5
45	5
90	13

Our aim is to train graph neural networks to predict the long-term kinetic characteristics of phase-separated systems, so we set up different datasets. Datasets 1 and 2 were both simulated with kinetic symmetry, and the rejection coefficients were 25 for the same particle type in dataset 1 and dataset 2, 50 for different particle types in dataset 1, and 25 for different particle types in dataset 2. To explore the effect of component ratios on the model,

we set up datasets 3 and 4 and chose different particle composition ratios to predict the position of particles. Dataset 3 is set to $N_A : N_B = 1 : 1$ and dataset 4 is $N_A : N_B = 4 : 1$.

The above datasets were simulated in Lammmps, and each configuration was run 200 times independently.

2.3 Order parameter design

We designed the order parameter to describe the local dynamic density, the box is partitioned into unit squares of independent size, and the global density distribution over time is calculated based on the cumulative number and species of particles in each square, the order parameter Φ is defined as follows.

$$\Phi = F \left\{ \sum_{i=1, j=1}^n (\varphi, t) \right\} \quad (4)$$

Where single area of $\varphi = |N_A - N_B| / (N_A + N_B)$, F is a normalized function in time, clearly $0 < \Phi < 1$.

2.4 Graph Neural Networks

Graph neural networks incorporating relational inference models have been shown to be a powerful way to discover hidden structures and physical laws from the evolution of physical locations²⁵⁻²⁷.

In our model, $G(\mathbf{V}, \mathbf{E})$ is a graph containing a set of nodes $\mathbf{V}$ and a set of edges $\mathbf{E}$, for a pair of points $u, v \in \{\mathbf{V}\}$ on the graph, and a connection $\mathbf{e}_{uv} \in \{\mathbf{E}\}$ between the two points. As shown in Fig.1, we take the type and location of the particles as input and set the sum of the cutoff radius and neighboring particle communication distances in the phase separation model to $k = 1.5$, and the particle pairs with distances on the edges less than this threshold (distances($\mathbf{e}_{uv}$) $< k$) are selected as the characteristic particle pairs. We use the particle types (different particle types are coded as 0 and 1) and particle positions of all particles at the initial moment as the features of points, the distance between all feature particle pairs and the conservative force (the repulsive force exerted by particles on each other in the direction along the edge) calculated based on this distance as the features of edges, and input all these features into GNN to form the graph data. Among them, the features of the edges will be first encoded and updated in the process of network update or by introducing two dynamic parameters (α and β) for combination, the network will automatically optimize these two parameters during the training process. And the features of the points of the particles will be encoded directly, and these input features will be updated n times by the MLPs, and the messages will be passed between the nodes and the edges, and the conserved forces between the pairs of feature particles as the features of the edges will be updated n times at the same time, and the decoder will generate the output features at the nodes and edges to finally predict the positions of the particles.

Considering the density of the simulated system as 3, we set the number of cyclic updates n to 5 in order to be able to traverse all particles within a threshold k . In addition to this, the encoder, MLP and decoder are implemented by three hidden layers with

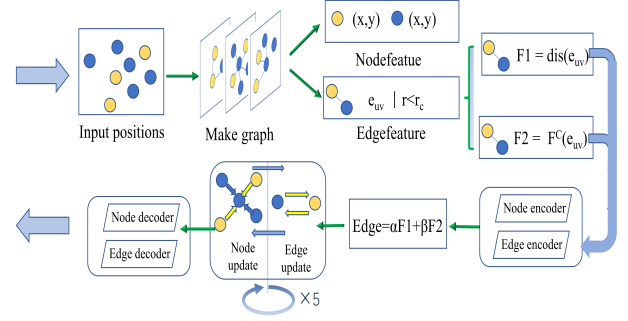


Fig. 1 graph neural network model to encode and decode features of points and features of edges, respectively, where the updating process is implemented by MLPs with two hidden layers.

64 linear units each and are updated nonlinearly by means of an activation function (ReLU). For the error between the predicted and true values, we set as the average difference between the positions of each particle at different moments.

$$loss = \frac{1}{N} \sum_{i=1}^n |X_i - X_{ruth}| + |Y_i - Y_{ruth}| \quad (5)$$

3 Experimental results

3.1 Description of phase separation order parameters

To begin with, we simulated the phase separation process in the two-dimensional system with $N_A : N_B = 1 : 1$, $a_{AB} = 25(A = B)$, $a_{AB} = 50(A \neq B)$. As shown in Fig.2, it can be seen that the evolution of phase separation becomes more and more stable as time changes Φ gradually from 0 to 1, while the A and B particles also keep gathering, and Φ remains relatively stable after $t = 4000$, which indicates that the order parameter Φ can describe the process of phase separation well.

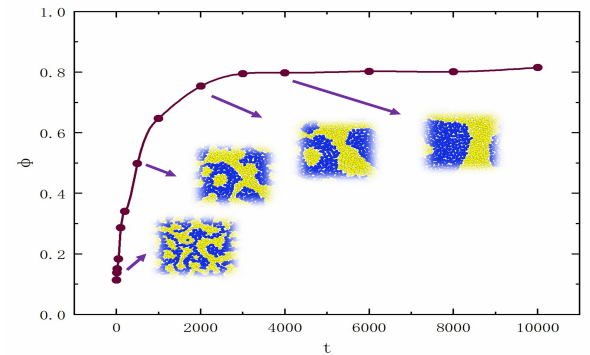


Fig. 2 The plot of the variation of the order parameter Φ with respect to time t . We took the data of 14 points in the simulation step.

3.1.1 Simulation of phase separation with different repulsion coefficients

In DPD simulations, the repulsion coefficients between different particles are the key factors driving the system into the phase separation state. We re-simulated the phase separation course at

different repulsion coefficients, chosen as 25, 36 and 50 (corresponding to the pure fluid, shallow quenched and deep quenched states, respectively), and compared the time evolution of the pure fluid with the morphological evolution of the binary mixture under deep and shallow quenching.

As shown in Fig.3(a), it can be seen that the value of Φ gradually approaches to 1 as the simulation time changes. For the pure fluid state, the value of Φ will finally stabilize around 0.12 ($t=6000$), when it can be observed that the A and B particles are completely mixed together and the phase separation state cannot be observed. In the shallow quenched state, the Φ value rises to about 0.63, which is different from 0.81 in the deep quenched state, but the two phases are already separated. Fig.3(b) represents the effect of the repulsion coefficient on the order parameter at $t = 6000$. Based on this variation, we verify three mechanisms of phase separation. Namely, in the case of $a_{AB} < 28$, the growth exponent is very low, after which Φ increases with the increase of the rejection coefficient. And after $a_{AB} > 40$, Φ is relatively smooth and reaches a stable state. We hope that the variation of Φ will provide a reasonable boundary between shallow and deep quench. As shown in Fig.3, the main reason for the difference is the formation of small islands of one fluid trapped by the other under shallow quenching, and the small islands of phase B remain trapped in phase A even in the later stages of shallow quenching phase separation due to the weak interaction between the same particles. In contrast, this is not the case in the deep quenching case.

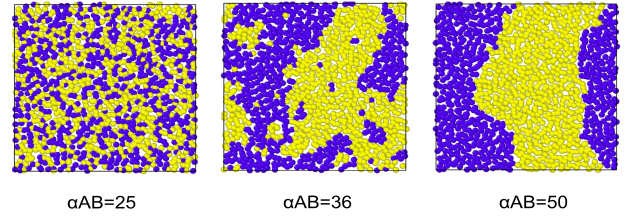
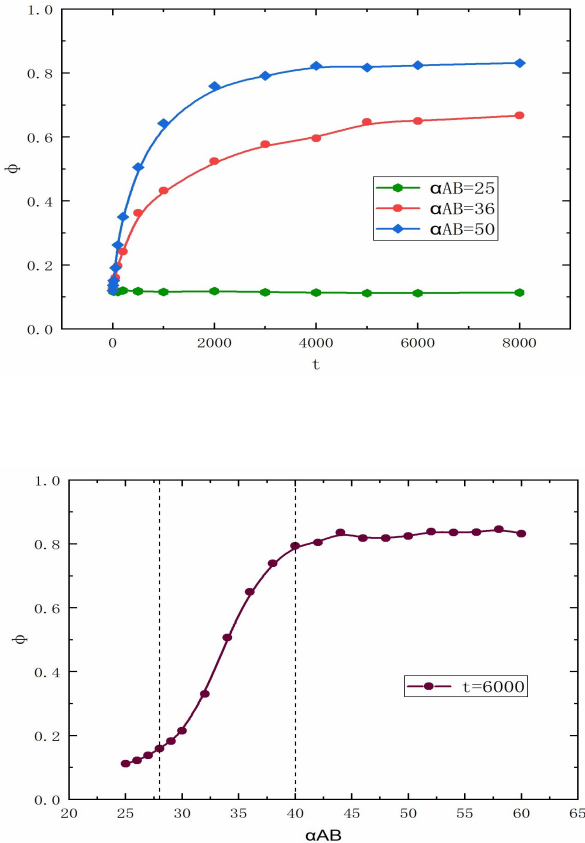


Fig. 3 (a) Variation of the local density order parameter Φ with simulation time t for different repulsion coefficients. (b) $t=6000$, the variation of the parameter Φ with respect to the repulsion coefficient, the two dashed lines correspond to $a_{AB} = 28$ and $a_{AB} = 40$. (c) Snapshots with repulsion coefficients of 25, 36, and 50 between different particles at $t = 4000$, respectively.

3.1.2 Simulation of phase separation for symmetric and asymmetric Schmidt numbers

In phase separation systems with binary fluids, the difference in viscosity of the different fluids greatly affects the dynamic behavior of the phase separation. The fluids in a multiphase system do not only interact with themselves but also with one another.

It is shown that the Schmidt number determines the relative importance of momentum and mass diffusion, and it plays an important role in the phase separation dynamics. Simulation parameters such as friction coefficient and cutoff radius can be adjusted to increase the Schmidt number and produce realistic interfacial effects on multiphase systems.

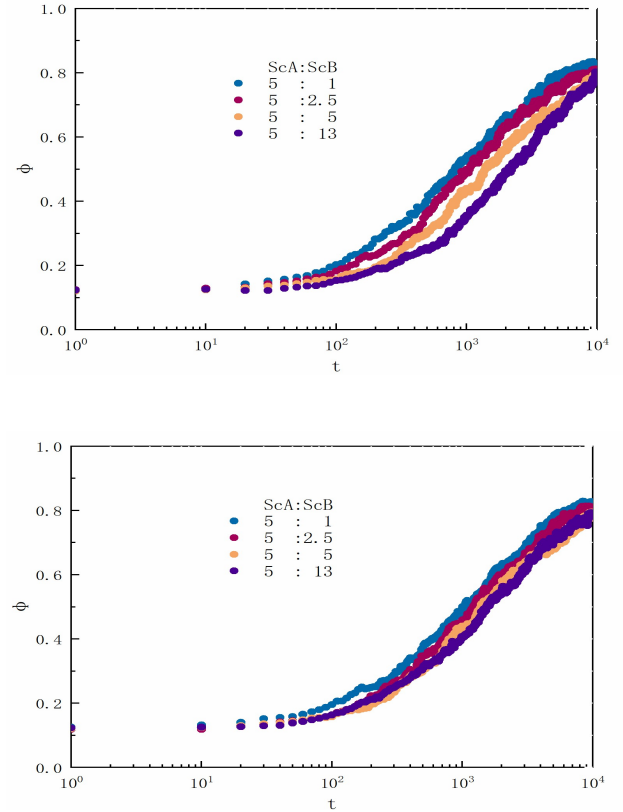


Fig. 4 (a) The variation of the order parameter Φ with time at symmetric Schmidt number. (b) The variation of the order parameter Φ with time at asymmetric Schmidt number.

As shown in Fig.4(a), Φ varies with respect to time for symmetric Schmidt number, and as the Schmidt number increases, the viscosity between the particles increases, which greatly affects the rate of phase separation. Although the rate of phase separation varies, the rate of phase separation slows down for systems with lower viscosity as time changes, and eventually all systems reach similar values of Φ .

Fig.4(b) indicates the variation of Φ with respect to time under the influence of asymmetric Schmidt number. It can be seen that the asymmetric Schmidt number accelerates the rate of phase separation, but still maintains the normal trend. This acceleration efficiency increases with asymmetry, and the system shows a higher separation rate when the viscosity of a component decreases relative to the system with symmetric Schmidt number. This is mainly due to the fact that the viscosity gap between the particles is large, and the reduction in viscosity allows the diffusion rate of the fluid to increase, which affects the transfer of kinetic effects because of the viscosity gap, making the connectivity between domains enhanced.

3.2 Phase separation prediction based on graph neural network

Machine learning has been proposed for learning the complexity and correlation between multidimensional data. The properties of machine learning in phase-separated systems are well suited to capture changes in the spatial position of particles, which also makes large-scale simulations a reality.

Here, we propose a new GNN model, an architecture suitable for extracting complex relationships from simulation data computed with high accuracy. It can require only the particle positions at the initial moment, form point and edge features by connecting neighboring particles, and continuously learn the changing forces during training. The whole interaction network consists of an encoder-process-decoder architecture, in which the features of nodes and edges in the graph can be computed mutually by exchanging messages between a pair of multilayer perceptrons (MLPs) assigned for nodes and edges.

3.2.1 Prediction of order parameters under different conditions

We perform the prediction of the local density order parameter for the phase separation system with different repulsion coefficients and different particle ratios. From Fig.5, it can be seen that the GNN model predicts Φ with lower accuracy in the shallow quenched state (repulsion coefficient between different particles is 30) than in the deep quenched (repulsion coefficient between different particles is equal to 50) state.

This is mainly due to the fact that the forces between the particles tend to be consistent, resulting in a bias towards disorderly particle motion. The conservative forces we choose as the characteristics of the edges, and this characteristic does not improve the model significantly when predicting the particle positions of shallow quenched systems. Besides, our model shows excellent performance as the simulation time varies, with an overall average error rate within 5%. This indicates that GNN is very sensitive to changes in local density, especially changes in distance due to

the interactions between different particles.

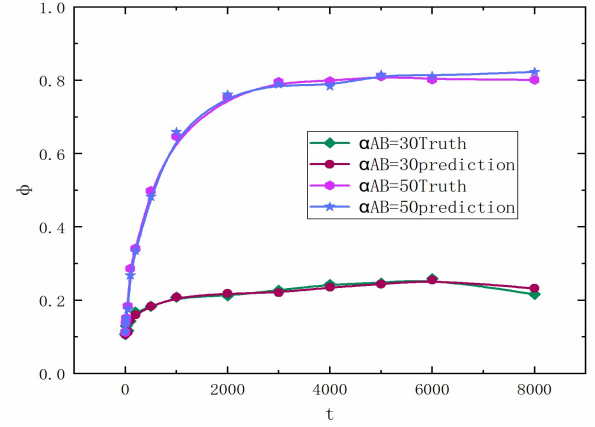


Fig. 5 The true and predicted values of the order parameter Φ for different repulsion coefficients.

3.2.2 Prediction of particle positions for different particle composition ratios

GNNs have been used as a powerful tool to discover hidden features from relational data. GNNs have unprecedented ability to predict particle propensity in predicting glass properties^{19,20}. Pearson correlation coefficients are often used to describe the correlation between data, between -1 and 1, with closer to 1 indicating closer proximity. We use the Pearson correlation coefficient to measure the accuracy between the predicted particle positions and the actual positions.

$$p = \sum_{i=1}^n \frac{\text{corrcoef}(x_{pre}, x_{mouth}) + \text{corrcoef}(y_{pre}, y_{truth})}{2}. \quad (6)$$

Fig.6 shows the prediction accuracy of our model regarding the position of the phase-separated particles. It can be seen that our model has a very high accuracy (Pearson correlation coefficients are all above 0.5) in the short term, especially in the time period when the phase separation evolves faster, and the simulation accuracy decreases as the time lengthens. Interestingly, the accuracy of the simulation is higher in the case of kinetic symmetry relative to kinetic asymmetry. This may be due to the fact that conservative forces generated by different types of particles in the characteristic particle pairs are stronger than those of the same type of particles, while in the state where the ratio of particle A to particle B is 4:1, there are more weak conservative forces between particles of the same type, which also makes the conservative forces as the characteristics of the edges more evenly distributed and more accurate in the case of kinetic symmetry. For the system studied so far, GNN shows significant prediction accuracy in the time range $t < 1000$. In fact, our model first learns the short-time dynamics on the edges and then gains the ability to predict the particle interaction forces on the nodes, which can be understood from the Pearson coefficients and the training duration.

Experiments show that GNN reproduces well the spatial distribution of local density variations; in particular, for short time

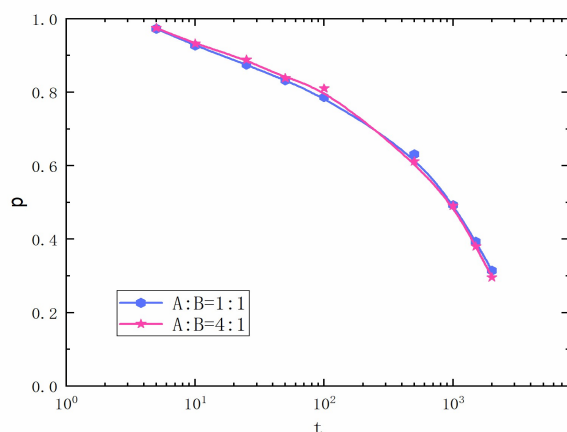


Fig. 6 Pearson correlation coefficients of particle positions with respect to the simulation time (we chose 10 points to predict).

predictions $t < 1000$, we find that the model is able to discriminate between specific force and distance variations produced by different types of particles. For long times $t > 1000$, the predictions become more diffuse, but still capture the contrast between regions of variation. Overall, GNN uses the information learned from the interaction forces and distance variations of paired particles to fix prediction errors over long time periods with high accuracy within the model.

4 Conclusions

In this work, we simulated the phase separation of binary mixtures under different conditions and measured the evolution of the phase separation by introducing an ordinal parameter of local density. The results show that our order parameter can effectively describe the process of two-phase separation and is validated for different systems. Besides, we design a data-driven GNN model based on the particle characteristics of the initial position only, and by learning the conserved forces and relative distances between the characteristic particle pairs, we successfully predict the sequence covariance changes in multiple states and make long-term predictions for the particle positions. Experiments demonstrate that our model has strong robustness and can be suitable for phase separation models in a variety of regimes.

The machine learning principle of our proposed model can be used for new multipurpose applications, as it can acquire the ability to predict long-term particle positions due to kinetic properties, which is of special interest for physical simulations. As a next step, we intend to introduce the GNN model within the 3D phase separation system, considering the phase order competition between multiple constituents and the phase separation process due to specific substances (colloids, block copolymers, etc.). Also adding specific structural and angular descriptors is something we have been wanting to do, which will facilitate the integration of data science and physical simulation even more.

Conflicts of interest

There are no conflicts to declare.

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